

Skills Summary

Comparison, Integral, Ratio, and Alternating-Series Tests

Use this reference while completing your worksheet or when studying.

Skill 1 — Limit comparison

Use it for terms that are a ratio of polynomials (or behave like one). Keep only the leading power on top and bottom to pick the comparison series b_n , then compute $L = \lim a_n/b_n$. If L is finite and positive, the two series live or die together.

Example: Test $\sum \frac{2n+1}{n^3+7}$ against $\sum \frac{1}{n^2}$.

Step 1. The terms are a ratio of polynomials, so limit comparison is the tool; leading powers give $n/n^3 = 1/n^2$ as the partner.

Step 2. $\frac{(2n+1)/(n^3+7)}{1/n^2} = \frac{2n^3+n^2}{n^3+7}$, whose limit is finite and positive.

$\Rightarrow L = 2$, and since $\sum 1/n^2$ converges, so does the series

Try it: Find L for $\sum \frac{5n-2}{n^3+1}$ against $\sum \frac{1}{n^2}$.

check: 5, so it converges

Skill 2 — Integral test

Use it when $a_n = f(n)$ for a function you can integrate. Check three things first: f positive, continuous, and decreasing on $[N, \infty)$. Then $\sum a_n$ and $\int_N^\infty f$ converge together. The integral is *not* the sum — it only decides the question.

Example: Test $\sum_{n=1}^\infty \frac{1}{n^2}$ with $f(x) = 1/x^2$.

Step 1. The terms come from a function that is easy to integrate and is positive, continuous and decreasing past $x = 1$, so the integral test applies.

Step 2. $\int_1^\infty x^{-2} dx = [-x^{-1}]_1^\infty = 0 + 1$, a finite value.

$\Rightarrow 1$, so the series converges

Try it: Evaluate $\int_1^\infty \frac{dx}{x^3}$ and say what it decides.

check: $\frac{2}{1}$, so it converges

Skill 3 — Ratio test

Use it for factorials and n th powers. Compute $L = \lim |a_{n+1}/a_n|$. Then $L < 1$ means convergence, $L > 1$ (including infinite) means divergence, and $L = 1$ means the test is silent — which is exactly what happens on rational terms, so do not start there.

Example: Test $\sum_{n=0}^{\infty} \frac{3^n}{n!}$.

Step 1. A factorial in the denominator makes consecutive terms easy to divide, so the ratio test is the natural first move.

Step 2. $\frac{a_{n+1}}{a_n} = \frac{3^{n+1}}{(n+1)!} \cdot \frac{n!}{3^n} = \frac{3}{n+1}$, and that shrinks to nothing.
 $\Rightarrow L = 0 < 1$, so the series converges

Try it: For $\sum \frac{n!}{2^n}$ the ratio is $\frac{n+1}{2}$. Find L and say what it decides.

check: ∞ , so it diverges

Skill 4 — Alternating series and its error bound

Use it when the signs alternate. Two conditions: the term sizes decrease, and their limit is 0. Then the series converges, and the error after N terms is at most the size of the *first term left out* — so the bound is a_{N+1} , read straight off the formula.

Example: For $\sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n^2}$, bound the error after 20 terms.

Step 1. Signs alternate and the sizes $1/n^2$ decrease to zero, so the remainder bound applies: find the first omitted term.

Step 2. Twenty terms run to $n = 20$, so the first omitted term is the one with $n = 21$, of size $1/21^2$.
 $\Rightarrow \frac{1}{441}$

Try it: For $\sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n^3}$, bound the error after 10 terms.

check: $\frac{1}{1331}$